

Geometric Realizations of L -Functions on the 600-Cell

(VFD programme — working draft)

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Abstract

We record explicit correspondences between sub-objects of the 600-cell / icosian ring and classical L -functions, together with their computational verification. Three correspondences are exhibited: (i) the icosian theta series over $K = \mathbb{Q}(\sqrt{5})$ realizes $\zeta_K(s)\zeta_K(s-1)$; (ii) the rational 24-cell sub-polytope realizes $\zeta(s)\zeta(s-1)$, whose zeros on the critical line are exactly the nontrivial zeros of the Riemann zeta function; and (iii) the Galois “twin” (the $\sqrt{5}$ -shell of 96 vertices) carries the quadratic Dirichlet L -function $L(s, \chi_5)$, in accordance with the classical factorization $\zeta_K = \zeta \cdot L(\chi_5)$. We state the scope precisely: these are *realizations and verifications of standard L -functions* attached to known lattices and number fields. **No claim is made about the Riemann Hypothesis, and no claim is made of a bridge to any physical model.**

1 Introduction and scope

The 600-cell is the regular 4-polytope whose 120 vertices are the unit icosians, i.e. the binary icosahedral group $2I \subset \mathbb{H}$. Its vertex set carries rich arithmetic: the icosian ring is a maximal order in a quaternion algebra over $K = \mathbb{Q}(\sqrt{5})$, and the associated lattice is a copy of E_8 [1]. This note isolates three *exact* arithmetic objects attached to 600-cell sub-structures and verifies, numerically, which L -functions they realize.

What is claimed. Only the correspondences in §3–§5 and their numerical verification (§6). All number-theoretic inputs (Dedekind factorization, functional equations) are standard and cited.

What is not claimed. Nothing about the location of the zeros of ζ (the Riemann Hypothesis); no positivity, operator, or Frobenius statement; no connection to any cosmological or physical model. These boundaries are deliberate and load-bearing.

2 Preliminaries

Let $K = \mathbb{Q}(\sqrt{5})$, $\varphi = \frac{1+\sqrt{5}}{2}$, $\mathcal{O}_K = \mathbb{Z}[\varphi]$. The Dedekind zeta function $\zeta_K(s)$ factors classically as

$$\zeta_K(s) = \zeta(s) L(s, \chi_5), \tag{1}$$

where χ_5 is the (even, real) quadratic Dirichlet character modulo 5 [2]. Both factors satisfy functional equations; ζ has a simple pole at $s = 1$, $L(\cdot, \chi_5)$ is entire.

The icosian ring $\mathcal{I}$ is the $\mathbb{Z}[\varphi]$ -order generated by the 120 unit icosians (the 600-cell vertices). Its 24-element rational sub-structure is the ring of Hurwitz units (the 24-cell / binary tetrahedral group $2T$), defined over $K = \mathbb{Q}$. The remaining 96 vertices form the φ -shell; the coordinate

Galois involution $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ fixes the 24-cell pointwise and permutes the 96 φ -vertices to the odd-permutation twin.

For a (positive-definite) lattice Λ we write $\vartheta_\Lambda(\tau) = \sum_{x \in \Lambda} q^{\langle x, x \rangle / 2}$, $q = e^{2\pi i \tau}$, for its classical theta series, and $L(\vartheta_\Lambda, s)$ for the standard L -function of the associated modular form. In §3, where the relevant form lives over $K = \mathbb{Q}(\sqrt{5})$, ϑ denotes instead the corresponding *Hilbert* theta series (over the two infinite places of K); we make this explicit there.

3 The icosian theta series and $\zeta_K(s)\zeta_K(s-1)$

The clean identity below rests on one arithmetic input—that the icosian order has a single class in its genus. We isolate this input as a lemma and pin it to the literature, since it is the load-bearing hypothesis of the section.

Lemma 1 (class number one). *Let B be the definite quaternion algebra over $K = \mathbb{Q}(\sqrt{5})$ ramified exactly at the two infinite places and unramified at all finite places, and let $\mathcal{I} \subset B$ be the icosian maximal order (the $\mathbb{Z}[\varphi]$ -order on the 120 unit icosians). Then $\mathcal{I}$ has class number one: a single left-ideal class.*

Reference. The pair $(B, \mathcal{I})$ is the maximal-order case with invariants $(n, d_F, D, N) = (2, 5, 1, 1)$ in the Kirschmer–Voight classification of definite quaternion orders of class number one over totally real fields [7, Table 8.2] (see also Vignéras [8]); it is the classical class-number-one example over $\mathbb{Q}(\sqrt{5})$. This is the left-ideal class number of the order, the quantity entering the Siegel–Weil average below. (The underlying $\mathbb{Z}$ -lattice of $\mathcal{I}$ is E_8 ; we note this only as context— E_8 ’s uniqueness as an even unimodular $\mathbb{Z}$ -lattice is a statement about $\mathbb{Z}$ -lattice genera, not about quaternion ideal classes, and is *not* used here.) $\square$

Proposition 2. *Let $\vartheta_{\mathcal{I}}$ be the theta series of the icosian order over $K = \mathbb{Q}(\sqrt{5})$. Then*

$$L(\vartheta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1).$$

Proof sketch. We take $\vartheta_{\mathcal{I}}$ to be the Hilbert theta series over K of the quaternionic norm form on $\mathcal{I}$ (weight two, the level determined by the discriminant $(D, N) = (1, 1)$), normalized so that its first nonzero Fourier coefficient is 1; and $L(\vartheta_{\mathcal{I}}, s)$ its standard (Hecke) L -function. By the Siegel–Weil formula [5, 6], the genus-average of such a theta series is the corresponding Eisenstein series. By Lemma 1 the order has a single ideal class, so the average reduces to the single term $\vartheta_{\mathcal{I}}$; hence $\vartheta_{\mathcal{I}}$ is that Eisenstein series, with no cuspidal contribution. The standard L -function of the relevant weight-two Hilbert Eisenstein series over K is the rank-two product $\zeta_K(s)\zeta_K(s-1)$ (constant-term / Hecke-eigenvalue computation; see [3, 4]). We present this as a *sketch*: the precise Eisenstein normalization and local factors are standard but not reproduced here, and the identity’s force in this note rests jointly on the sketch and the numerical corroboration below. $\square$

Remark 3. *Class number one is what makes the identity clean: were the order to have $h > 1$ ideal classes, $\vartheta_{\mathcal{I}}$ would decompose as the Eisenstein series plus a cusp form, and $L(\vartheta_{\mathcal{I}}, s)$ would be a sum carrying additional cuspidal L -contributions rather than the bare product $\zeta_K(s)\zeta_K(s-1)$. Lemma 1 excludes this.*

Remark 4 (numerical corroboration; provenance). *The identity was corroborated numerically in the companion icosian/Brandt computation of this programme (the **icosian-triad** work), where the representation numbers $r(\pi)$ were obtained by exact short-vector enumeration over $\mathbb{Z}[\varphi]$ and matched, out of sample, against the Dirichlet coefficients of $\zeta_K(s)\zeta_K(s-1)$. We flag provenance*

honestly: the verification-engine scripts shipped with this bundle (§6, `vfd_math_engine`) illustrate the gate by declaring the predicted zero lists directly; they do not re-derive $r(\pi)$ from the lattice. The geometry→arithmetic enumeration is the separate computation just cited.

4 The rational 24-cell and $\zeta(s)\zeta(s-1)$

Applying the same construction to the rational sub-order (the 24-cell, $K = \mathbb{Q}$) yields the $\mathbb{Q}$ -analogue of Result 2.

Computational Result 5. The 24-cell (Hurwitz) theta series realizes

$$L(\vartheta_{24}, s) = \zeta(s) \zeta(s-1).$$

Proposition 6. *On the critical line $\operatorname{Re}(s) = \frac{1}{2}$, the zeros of $\zeta(s)\zeta(s-1)$ are exactly those nontrivial zeros of ζ that lie on that line. (Equivalently: the factor $\zeta(s-1)$ contributes none of them.)*

Proof. A nontrivial zero of $\zeta(s-1)$ occurs at $s-1 = \rho$, i.e. $s = 1 + \rho$; since every nontrivial zero ρ has $\operatorname{Re}(\rho) \in (0, 1)$, such s has $\operatorname{Re}(s) \in (1, 2)$, never $\frac{1}{2}$. The trivial zeros of $\zeta(s-1)$ lie at $s = 1 - 2n$ ($n \geq 1$), also off the line. Hence on $\operatorname{Re}(s) = \frac{1}{2}$ the factor $\zeta(s-1)$ contributes no zeros, and the on-line zeros of the product are precisely the on-line zeros of $\zeta(s)$. This is a statement about *which* zeros lie on the line, and asserts nothing about whether all nontrivial zeros of ζ do (see §7). $\square$

Thus the 24-cell — the $\sqrt{5}$ -free, Galois-fixed sub-polytope — carries the Riemann zeros on its critical line, with the second factor displaced to $\operatorname{Re}(s) = \frac{3}{2}$. (This is a statement about *which* zeros lie on the line, not about whether all zeros of ζ do; see §7.)

5 The Galois twin and $L(s, \chi_5)$

The factorization (1) predicts that the “ $\sqrt{5}$ part” of the icosian structure carries $L(\chi_5)$. Geometrically, this is the φ -shell.

Computational Result 7. The 96-vertex φ -shell (the Galois-sign component) realizes the Dirichlet L -function $L(s, \chi_5)$: its on-line zeros coincide (to tested precision) with the zeros of $L(s, \chi_5)$; no coincidence with a Riemann zero was observed in the tested range.

Together, Results 2–7 are mutually consistent via (1): the full icosian object carries $\zeta_K(s)\zeta_K(s-1) = \zeta(s)L(s, \chi_5)\zeta(s-1)L(s-1, \chi_5)$; its Galois-fixed part carries the ζ -factor (the 24-cell), and its Galois-sign part carries the $L(s, \chi_5)$ factor (the φ -shell).

6 Verification

The correspondences were checked computationally. For each claimed L -function we compared its on-line zeros against the target’s zeros on four invariants: (a) first- k zero positions; (b) zero density (the Weyl/Riemann–von Mangoldt rate); (c) the nearest-neighbour spacing distribution (“fp” below is the fraction of normalized gaps below 0.3); (d) the number variance $\Sigma^2(\ell)$. The geometry→arithmetic input—the icosian representation numbers $r(\pi)$ by exact short-vector enumeration over $\mathbb{Z}[\varphi]$, matched out of sample against $\zeta_K(s)\zeta_K(s-1)$ —was performed in the companion icosian/Brandt computation (cited in §3). The engine scripts in *this* bundle exercise the verification gate on declared zero lists and are not the enumeration; the figures below are reproduced from the cited computation.

correspondence (on-line content)	first on-line zeros	spacing fp (< 0.3)
24-cell $\rightarrow \zeta(s)\zeta(s-1)$	14.13, 21.02, 25.01, 30.42	0.000
φ -shell $\rightarrow L(s, \chi_5)$	6.65, 9.83, 11.96, 16.03	0.000
icosian $\rightarrow \zeta_K(s)\zeta_K(s-1)$	6.65, 9.83, 11.96, 14.13	0.077

The single- L -function rows are consistent with GUE nearest-neighbour repulsion ($\text{fp} \approx 0$); the rank-four icosian row shows the larger value expected for a *superposition* of independent spectra ($\zeta_K(s)\zeta_K(s-1) = \zeta(s)L(s, \chi_5)\zeta(s-1)L(s-1, \chi_5)$), consistent with (1). These are diagnostic comparisons against the named target L -functions, not independent evidence for the geometric attachment.

7 Scope and what is not claimed

We restate the boundary. Propositions 6 and Results 2–7 describe *which* L -functions are realized by which 600-cell sub-objects, and which of their zeros lie on the critical line. They are **silent on the Riemann Hypothesis**: the claim “the 24-cell carries the Riemann zeros” refers to the zeros that *do* lie on $\text{Re}(s) = \frac{1}{2}$, and asserts nothing about whether *all* nontrivial zeros of ζ lie there. No positivity, self-adjointness, or arithmetic-surface statement is made or implied. A companion note records, with explicit certificates, that the natural “substrate $\leftrightarrow \zeta$ ” correspondence is *negative* — the icosian object realizes $\zeta_K(s)\zeta_K(s-1) = \zeta(s)L(s, \chi_5)\zeta(s-1)L(s-1, \chi_5)$, which contains ζ only as a factor and does not isolate it.

References

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